Inequalities — problem 7.89

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Let x, y, z be positive reals with $xy + yz + zx = 1$. Prove the inequality

$$\frac{2x(1-x^2)}{(1+x^2)^2} + \frac{2y(1-y^2)}{(1+y^2)^2} + \frac{2z(1-z^2)}{(1+z^2)^2} \leq \frac{x}{1+x^2} + \frac{y}{1+y^2} + \frac{z}{1+z^2}.$$